AN APPROXIMATE PRESENTATION WITH DIVISORIAL JACOBIAN BOUND ALGEBRAIZES TO THE COMPLETION OF A FINITE TYPE ALGEBRA

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a rig-smooth reduction of the hypotheses, an engineered finite type algebra whose Jacobian determinant gives a truncation-independent annihilation bound, a lift of the mod-I^n presentation to a genuine surjection, and a kernel control finished by a free-module rank computation after inverting b

Editorial note. The label lemma-approximate-presentation-rig-smooth names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The category of topological A-algebras complete along I for a Noetherian ring A with ideal I (the displayed category in the source section), and the rig-smooth property over the pair (A, I)
- The completed naive cotangent complex NL and its long exact sequence, as developed in the same chapter
- The uniformity integers d and q of the associated graded Gr_I(B) from the local cohomology section cited in the proof

Source reference. Original lemma 0315: sources/stacks-chow/tags.txt, source lines 3972–3972. Prerequisite pointer invoked at proof line(s) 1726; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 0AL4: sources/stacks-restricted/pinned-original-source.tex, source lines 326–326. Prerequisite pointer invoked at proof line(s) 1640; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GAF: sources/stacks-restricted/pinned-original-source.tex, source lines 560–560. Prerequisite pointer invoked at proof line(s) 1697; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GAJ: sources/stacks-restricted/pinned-original-source.tex, source lines 843–843. Prerequisite pointer invoked at proof line(s) 1657;

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it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 00S2: sources/stacks-restricted/pinned-original-source.tex, source lines 725–725. Prerequisite pointer invoked at proof line(s) 1744; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GAQ: sources/stacks-restricted/pinned-original-source.tex, source lines 1167–1167. Prerequisite pointer invoked at proof line(s) 1718; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GAV: sources/stacks-restricted/pinned-original-source.tex, source lines 1598–1598. Prerequisite pointer invoked at proof line(s) 1674; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0GA6: sources/stacks-chow/tags.txt, source lines 20326–20326. Prerequisite pointer invoked at proof line(s) 1717; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0G9H: sources/stacks-chow/tags.txt, source lines 20302–20302. Prerequisite pointer invoked at proof line(s) 1701; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0ALR: sources/stacks-chow/tags.txt, source lines 13379–13379. Prerequisite pointer invoked at proof line(s) 1729; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0G9G: sources/stacks-chow/tags.txt, source lines 20301–20301. Prerequisite pointer invoked at proof line(s) 1702; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let I be an ideal of a Noetherian ring A. Let B be an object of (0AL4). Let $B = A[x_1, \ldots, x_r]^\wedge/J$ be a presentation. Assume there exists an element $b \in B$, $0 \leq m \leq r$, and $f_1, \ldots, f_m \in J$ such that*

1. *$V(b) \subset V(IB)$ in $\mathrm{Spec}(B)$,*
2. *the image of $\Delta = \det_{1 \leq i,j \leq m}(\partial f_j/\partial x_i)$ in B divides b, and*
3. *$bJ \subset (f_1, \ldots, f_m) + J^2$.*

Then there exists a finite type A-algebra C and an A-algebra isomorphism $B \cong C^\wedge$.

Proof. The conditions imply that B is rig-smooth over (A, I), see Lemma 0GAJ. Write $b'\Delta = b$ in B for some $b' \in B$. Say $I = (a_1, \ldots, a_t)$. Since $V(b) \subset V(IB)$ there exists an integer $c \geq 0$ such that $I^c B \subset bB$. Write $bb_i = a_i^c$ in B for some $b_i \in B$.

Choose an integer $n \gg 0$ (we will see later how large). Choose polynomials $f'_1, \ldots, f'_m \in A[x_1, \ldots, x_r]$ such that $f_i - f'_i \in I^n A[x_1, \ldots, x_r]^\wedge$. We set $\Delta' =$

$\det_{1\leq i,j\leq m}(\partial f'_j/\partial x_i)$ and we consider the finite type A-algebra

$$C = A[x_1, \ldots, x_r, z_1, \ldots, z_t]/(f'_1, \ldots, f'_m, z_1\Delta' - a_1^c, \ldots, z_t\Delta' - a_t^c)$$

We will apply Lemma 0GAV to C. We compute

$$\det \begin{pmatrix} \text{matrix of partials of} \\ f'_1, \ldots, f'_m, z_1\Delta' - a_1^c, \ldots, z_t\Delta' - a_t^c \\ \text{with respect to the variables} \\ x_1, \ldots, x_m, z_1, \ldots, z_t \end{pmatrix} = (\Delta')^{t+1}$$

Hence we see that $\mathrm{Ext}^1_C(NL_{C/A}, N)$ is annihilated by $(\Delta')^{t+1}$ for all C-modules N. Since a_i^c is divisible by Δ' in C we see that $a_i^{(t+1)c}$ annihilates these Ext^1's also. Thus I^{c_1} annihilates $\mathrm{Ext}^1_C(NL_{C/A}, N)$ for all C-modules N where $c_1 = 1 + t((t+1)c - 1)$. The exact value of c_1 doesn't matter for the rest of the argument; what matters is that it is independent of n.

Since $NL^\wedge_{C^\wedge/A} = NL_{C/A} \otimes_C C^\wedge$ by Lemma 0GAF we conclude that multiplication by I^{c_1} is zero on $\mathrm{Ext}^1_{C^\wedge}(NL^\wedge_{C^\wedge/A}, N)$ for any $C^\wedge$-module N as well, see More on Algebra, Lemmas 0G9H and 0G9G. In particular $C^\wedge$ is rig-smooth over (A, I).

Observe that we have a surjective A-algebra homomorphism

$$\psi_n : C \longrightarrow B/I^nB$$

sending the class of x_i to the class of x_i and sending the class of z_i to the class of b_ib'. This works because of our choices of b' and b_i in the first paragraph of the proof.

Let $d = d(\mathrm{Gr}_I(B))$ and $q_0 = q(\mathrm{Gr}_I(B))$ be the integers found in Local Cohomology, Section 0GA6. By Lemma 0GAQ if we take $n \geq \max(q_0 + (d+1)c_1, 2(d+1)c_1 + 1)$ we can find a homomorphism $\varphi : C^\wedge \to B$ of A-algebras which is congruent to ψ_n modulo $I^{n-(d+1)c_1}B$.

Since $\varphi : C^\wedge \to B$ is surjective modulo I we see that it is surjective (for example use Algebra, Lemma 0315). To finish the proof it suffices to show that $\mathrm{Ker}(\varphi)/\mathrm{Ker}(\varphi)^2$ is annihilated by a power of I, see More on Algebra, Lemma 0ALR.

Since φ is surjective we see that $NL^\wedge_{B/C^\wedge}$ has cohomology modules $H^0(NL^\wedge_{B/C^\wedge}) = 0$ and $H^{-1}(NL^\wedge_{B/C^\wedge}) = \mathrm{Ker}(\varphi)/\mathrm{Ker}(\varphi)^2$. We have an exact sequence

$$H^{-1}(NL^\wedge_{C^\wedge/A} \otimes_{C^\wedge} B) \to H^{-1}(NL^\wedge_{B/A}) \to H^{-1}(NL^\wedge_{B/C^\wedge}) \to H^0(NL^\wedge_{C^\wedge/A} \otimes_{C^\wedge} B) \to H^0(NL^\wedge_{B/A}) \to 0$$

by Lemma 00S2. The first two modules are annihilated by a power of I as B and $C^\wedge$ are rig-smooth over (A, I). Hence it suffices to show that the kernel of the surjective map $H^0(NL^\wedge_{C^\wedge/A} \otimes_{C^\wedge} B) \to H^0(NL^\wedge_{B/A})$ is annihilated by a power of I. For this it suffices to show that it is annihilated by a power of b. In other words, it suffices to show that

$$H^0(NL^\wedge_{C^\wedge/A}) \otimes_{C^\wedge} B[1/b] \longrightarrow H^0(NL^\wedge_{B/A}) \otimes_B B[1/b]$$

is an isomorphism. However, both are free $B[1/b]$ modules of rank $r - m$ with basis $\mathrm{d}x_{m+1}, \ldots, \mathrm{d}x_r$ and we conclude the proof. $\square$